a: Generalized Schwarzschild-de Sitter Metric

Starting Metric: $g_{00}=A(t,r)$, $g_{11}=-B(r)$, $g_{22}=-r^2$, $g_{33}=-r^2\sin^2\theta$

Line Element:

$$ds^2 = A[t, r]\, dt^2 - B[r]\, dr^2 - r^2\, d\theta^2 - r^2\, \text{Sin}[\theta]^2\, d\phi^2$$

With this metric, the non-zero field equations are:

$$R_{00}: \quad -\frac{A'}{A\, r\, B} + \frac{(A')^2}{4\, A^2\, B} + \frac{A'\, B'}{4\, A\, B^2} - \frac{A''}{2\, A\, B} = \Lambda$$

$$R_{11}: \quad \frac{(A')^2}{4\, A^2\, B} + \frac{B'}{r\, B^2} + \frac{A'\, B'}{4\, A\, B^2} - \frac{A''}{2\, A\, B} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{1}{r^2\, B} - \frac{A'}{2\, A\, r\, B} + \frac{B'}{2\, r\, B^2} = \Lambda$$

$$R_{33}: \quad \frac{1}{r^2} - \frac{1}{r^2\, B} - \frac{A'}{2\, A\, r\, B} + \frac{B'}{2\, r\, B^2} = \Lambda$$

We see that R_{33} is redundant.

$R_{00}-R_{11}-2R_{22}$ can be solved for $B(r)$;

$$R_{00}-R_{11}-2R_{22}: \quad -\frac{2}{r^2} + \frac{2}{r^2\, B} - \frac{2\, B'}{r\, B^2} = -2\, \Lambda$$

$$\Rightarrow \quad B(r) = \frac{1}{1 - \frac{\alpha}{r} - \frac{r^2\, \Lambda}{3}} \quad \text{where } \alpha \text{ is a constant of integration.}$$

With this formula for $B(r)$, the equations become

$$R_{00}: \quad -\frac{A'}{A\, r} + \frac{3\, \alpha\, A'}{4\, A\, r^2} + \frac{r\, \Lambda\, A'}{2\, A} + \frac{(A')^2}{4\, A^2} - \frac{\alpha\, (A')^2}{4\, A^2\, r} - \frac{r^2\, \Lambda\, (A')^2}{12\, A^2} - \frac{A''}{2\, A} + \frac{\alpha\, A''}{2\, A\, r} + \frac{r^2\, \Lambda\, A''}{6\, A} = \Lambda$$

$$R_{11}: \quad -\frac{\alpha}{r^3} + \frac{2\, \Lambda}{3} - \frac{\alpha\, A'}{4\, A\, r^2} + \frac{r\, \Lambda\, A'}{6\, A} + \frac{(A')^2}{4\, A^2} - \frac{\alpha\, (A')^2}{4\, A^2\, r} - \frac{r^2\, \Lambda\, (A')^2}{12\, A^2} - \frac{A''}{2\, A} + \frac{\alpha\, A''}{2\, A\, r} + \frac{r^2\, \Lambda\, A''}{6\, A} = \Lambda$$

$$R_{22}: \quad \frac{\alpha}{2\, r^3} + \frac{2\, \Lambda}{3} - \frac{A'}{2\, A\, r} + \frac{\alpha\, A'}{2\, A\, r^2} + \frac{r\, \Lambda\, A'}{6\, A} = \Lambda$$

Solving R_{22} for $A(t,r)$ yields:

$$A(t,r) = \left(1 - \frac{\alpha}{r} - \frac{r^2\, \Lambda}{3}\right) \mathcal{A}[t] = \frac{\mathcal{A}(t)}{B(r)}, \quad \text{where } \mathcal{A}(t) \text{ is a new unknown.}$$

Plugging this in, R_{22} becomes an identity and the other equations become:

R_{00}: $\Lambda = \Lambda$

R_{11}: $\Lambda = \Lambda$

All of the equations are satisfied,
and we have an exact solution to the Λ field equations:

$$A(t,r) = \frac{\mathcal{A}(t)}{B(r)}$$ where $\mathcal{A}(t)$ is a free parameter,

$$B(r) = \frac{1}{1 - \frac{\alpha}{r} - \frac{r^2 \Lambda}{3}}$$ where α is a free constant,

and line element

$$ds^2 = -\frac{\left(-3r + 3\alpha + r^3 \Lambda\right) \mathcal{A}[t]}{3r} dt^2 - \frac{1}{1 - \frac{\alpha}{r} - \frac{r^2 \Lambda}{3}} dr^2 - r^2 d\theta^2 - r^2 \text{Sin}[\theta]^2 d\phi^2$$

I call this the **Generalized Schwarzschild-de Sitter equations**.

Note: Just as free $C(r)$ is simply a relabeling of r in FG-GRW, free $\mathcal{A}(t)$ is simply a relabeling of t. This solution is not a new spacetime. This is Birkhoff's theorem: every spherically symmetric vacuum solution is static.

b: Schwarzschild-de Sitter Metric

The **Schwarzschild-de Sitter solution** is obtained by setting $\mathcal{A}(t) = c^2$ and $\alpha = \frac{2GM}{c^2}$:

$$\mathcal{A}(t) = c^2$$

$$B(r) = \frac{1}{1 - \frac{2GM}{c^2 r} - \frac{r^2 \Lambda}{3}}.$$

c: Schwarzschild Metric

The Schwarzschild solution is obtained by setting $\Lambda = 0$ in the Schwarzschild-de Sitter equations:

$$\mathcal{A}(t) = c^2$$

$$B(r) = \frac{1}{1 - \frac{2GM}{c^2 r}}.$$

d: de Sitter Metrics

The de Sitter and Anti de Sitter solutions are obtained by setting M = 0 in the Schwarzschild-de Sitter equations:

$$\mathcal{A}(t) = c^2$$

$$B(r) = \frac{1}{1 - \frac{r^2 \Lambda}{3}}.$$

When $\Lambda > 0$, these are the de Sitter equations. When $\Lambda < 0$, these are the Anti de Sitter equations.